Q1)Analyze the iris dataset. 1.Load the Iris Dataset

2.Calculate mean of each numerical feature.

3.Calculate median of each numerical feature.

4.Calculate mode(s) of each numerical feature.(Note:Mode can be multi-modal or cannot be calculated for continuous data)

5.Calculate standard deviation of each numerical feature.

6.Calculate variance of each numerical feature.

7.Calculate skewness of each numerical feature.

8.Calculate kurtosis of each numerical feature.

Also interpret the results

```
import pandas as pd
import numpy as np
import seaborn as sns
from scipy.stats import skew,kurtosis
import matplotlib.pyplot as plt
```

```
iris = sns.load_dataset('iris')
print(iris.head())
iris.info()
```

```
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
numerical_cols = iris.select_dtypes(include=np.number).columns.to_list()

for col in numerical_cols:
  mean_val = iris[col].mean()
  median_val = iris[col].median()
  mode_val = iris[col].mode().tolist()
  std_val = iris[col].std()
  var_val = iris[col].var()
  skewness_val = skew(iris[col])
  kurtosis_val = kurtosis(iris[col])

  print(f"---Stats for {col}---")
  print(f" Mean: {mean_val}")
  print(f" Median: {median_val}")
  print(f" Mode: {mode_val}")
  print(f" Standard Deviation: {std_val}")
  print(f" Variance: {var_val}")
  print(f" Skewness: {skewness_val}")
  print(f" Kurtosis: {kurtosis_val}")
  print("\n")
```

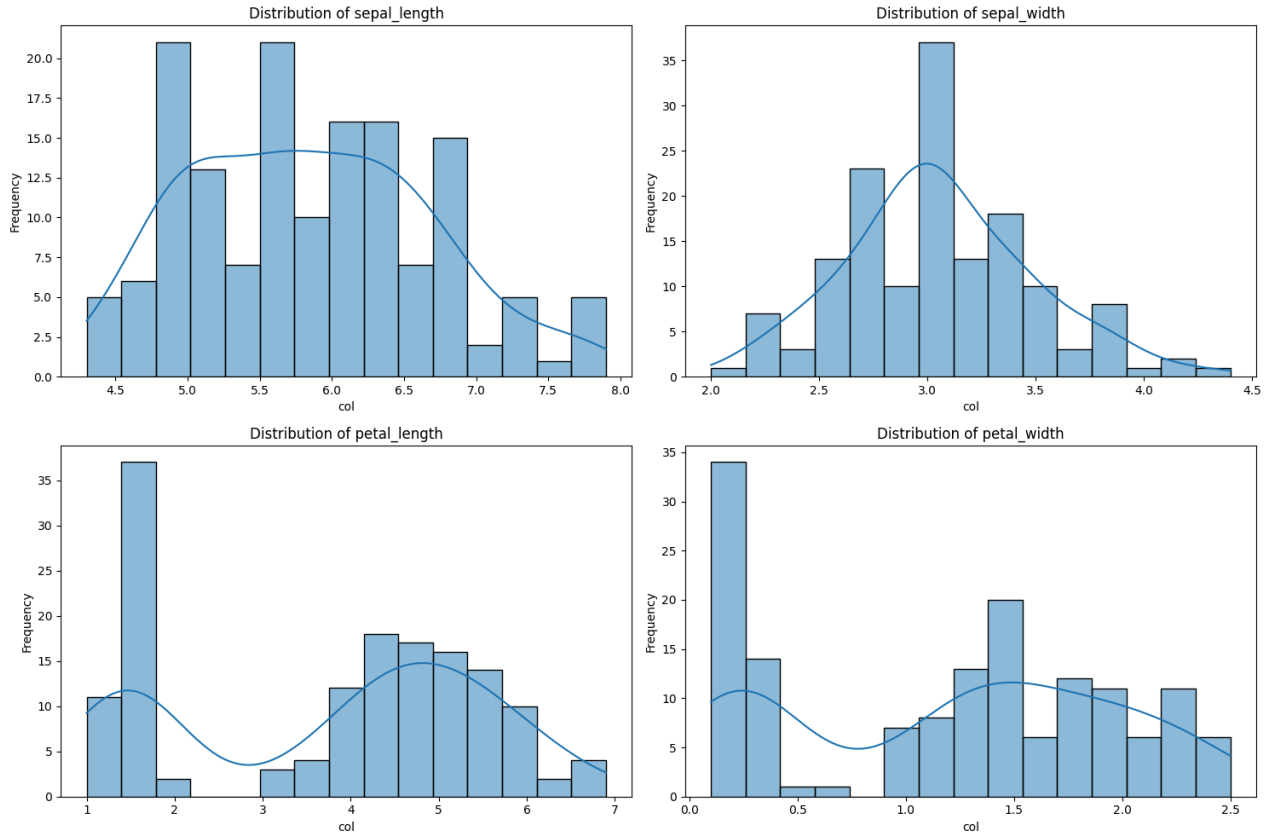
```
---Stats for sepal_length---
 Mean: 5.843333333333334
 Median: 5.8
 Mode: [5.0]
 Standard Deviation: 0.8280661279778629
 Variance: 0.6856935123042505
 Skewness: 0.3117530585022963
 Kurtosis: -0.5735679489249765


---Stats for sepal_width---
 Mean: 3.0573333333333337
 Median: 3.0
 Mode: [3.0]
 Standard Deviation: 0.435866284936698
 Variance: 0.1899794183445188
 Skewness: 0.31576710633893473
 Kurtosis: 0.18097631752246768
```

```
---Stats for petal_length---
 Mean: 3.7580000000000005
 Median: 4.35
 Mode: [1.4, 1.5]
 Standard Deviation: 1.7652982332594667
 Variance: 3.1162778523489942
 Skewness: -0.2721276664567214
 Kurtosis: -1.3955358863990055


---Stats for petal_width---
 Mean: 1.1993333333333336
 Median: 1.3
 Mode: [0.2]
 Standard Deviation: 0.7622376689603465
 Variance: 0.5810062639821029
 Skewness: -0.10193420656560036
 Kurtosis: -1.336067405231553
```

```
plt.figure(figsize=(15,10))
for i, col in enumerate(numerical_cols):
  plt.subplot(2,2,i+1)
  sns.histplot(iris[col],kde=True,bins=15)
  plt.title(f'Distribution of {col}')
  plt.xlabel('col')
  plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```



Interpretations:

Q2. Analyze the Tips dataset.

```
tips = sns.load_dataset('tips')
print(tips.head())
tips.info()
```

```
   total_bill   tip     sex smoker  day    time  size
0       16.99  1.01  Female     No  Sun  Dinner     2
1       10.34  1.66    Male     No  Sun  Dinner     3
2       21.01  3.50    Male     No  Sun  Dinner     3
3       23.68  3.31    Male     No  Sun  Dinner     2
4       24.59  3.61  Female     No  Sun  Dinner     4
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   total_bill  244 non-null    float64
 1   tip         244 non-null    float64
 2   sex         244 non-null    category
 3   smoker      244 non-null    category
 4   day         244 non-null    category
 5   time        244 non-null    category
 6   size        244 non-null    int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
```

```
numerical_cols = tips.select_dtypes(include=np.number).columns.to_list()

for col in numerical_cols:
  mean_val = tips[col].mean()
  median_val = tips[col].median()
  mode_val = tips[col].mode().tolist()
  std_val = tips[col].std()
  var_val = tips[col].var()
  skewness_val = skew(tips[col])
  kurtosis_val = kurtosis(tips[col])

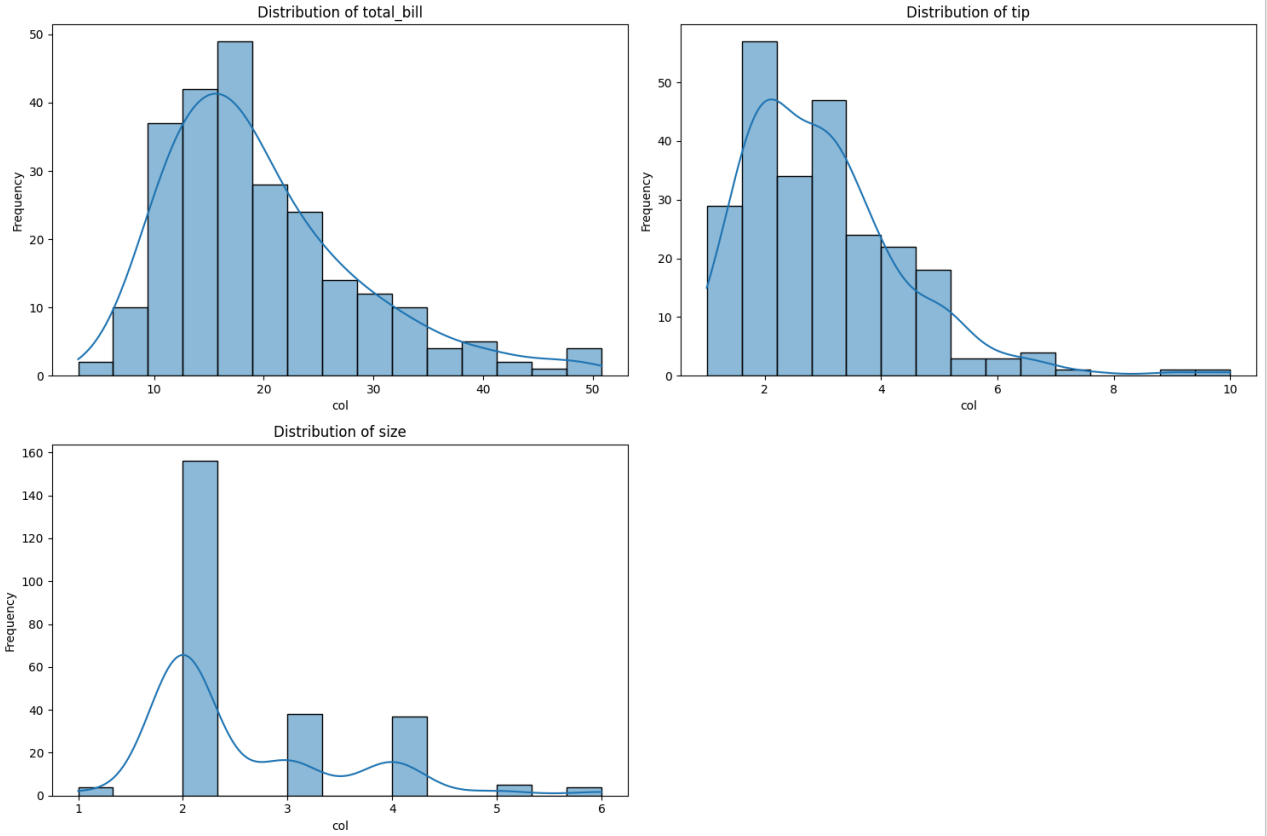
  print(f"---Stats for {col}---")
  print(f" Mean: {mean_val}")
  print(f" Median: {median_val}")
  print(f" Mode: {mode_val}")
  print(f" Standard Deviation: {std_val}")
  print(f" Variance: {var_val}")
  print(f" Skewness: {skewness_val}")
  print(f" Kurtosis: {kurtosis_val}")
  print("\n")
```

```
---Stats for total_bill---
 Mean: 19.78594262295082
 Median: 17.795
 Mode: [13.42]
 Standard Deviation: 8.902411954856856
 Variance: 79.25293861397826
 Skewness: 1.1262346334818638
 Kurtosis: 1.1691681323851366


---Stats for tip---
 Mean: 2.99827868852459
 Median: 2.9
 Mode: [2.0]
 Standard Deviation: 1.3836381890011826
 Variance: 1.9144546380624725
 Skewness: 1.4564266884221506
 Kurtosis: 3.5495519893455114


---Stats for size---
 Mean: 2.569672131147541
 Median: 2.0
 Mode: [2]
 Standard Deviation: 0.9510998047322332
 Variance: 0.9045908385616921
 Skewness: 1.4389653841920984
 Kurtosis: 1.6719276263625504
```

```
plt.figure(figsize=(15,10))
for i, col in enumerate(numerical_cols):
  plt.subplot(2,2,i+1)
  sns.histplot(tips[col],kde=True,bins=15)
  plt.title(f'Distribution of {col}')
  plt.xlabel('col')
  plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```



Interpretations:

```
import pandas as pd
stock = pd.read_csv('indian_stock_market.csv')
print(stock.head())
stock.info()
```

```
        MSFT        IBM       SBUX       AAPL         GSPC        Date
0  23.950705  80.517962  16.149666  11.086612  1416.599976  2007-01-03
1  23.910599  81.378851  16.167992  11.332687  1418.339966  2007-01-04
2  23.774242  80.642129  16.099269  11.251984  1409.709961  2007-01-05
3  24.006852  81.867244  16.039710  11.307550  1412.839966  2007-01-08
4  24.030914  82.835742  15.970989  12.246870  1412.109985  2007-01-09
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2306 entries, 0 to 2305
Data columns (total 6 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   MSFT    2306 non-null   float64
 1   IBM     2306 non-null   float64
 2   SBUX    2306 non-null   float64
 3   AAPL    2306 non-null   float64
 4   GSPC    2306 non-null   float64
 5   Date    2306 non-null   object
dtypes: float64(5), object(1)
memory usage: 108.2+ KB
```

```
numerical_cols = stock.select_dtypes(include=np.number).columns.to_list()

for col in numerical_cols:
  mean_val = stock[col].mean()
  median_val = stock[col].median()
  mode_val = stock[col].mode().tolist()
  std_val = stock[col].std()
  var_val = stock[col].var()
  skewness_val = skew(stock[col])
  kurtosis_val = kurtosis(stock[col])

  print(f"---Stats for {col}---")
  print(f" Mean: {mean_val}")
  print(f" Median: {median_val}")
```

```
  print(f" Mode: {mode_val}")
  print(f" Standard Deviation: {std_val}")
  print(f" Variance: {var_val}")
  print(f" Skewness: {skewness_val}")
  print(f" Kurtosis: {kurtosis_val}")
  print("\n")
```

```
---Stats for MSFT---
 Mean: 28.737785089332178
 Median: 24.990737000000003
 Mode: [20.7885, 22.432038, 23.442524, 23.780523, 23.799136, 23.812836, 24.375044, 24.883749, 26.236076, 27.52491, 27.622691
 Standard Deviation: 9.47173720879108
 Variance: 89.71380575239743
 Skewness: 1.0208522738773171
 Kurtosis: 0.11462575264900465

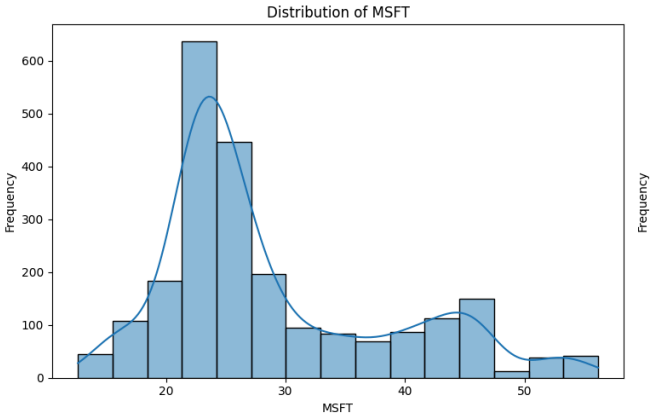

---Stats for IBM---
 Mean: 136.23959955550737
 Median: 143.8331255
 Mode: [180.597103]
 Standard Deviation: 37.37635951034845
 Variance: 1396.9922502468153
 Skewness: -0.14935923700494055
 Kurtosis: -1.4100218574371501

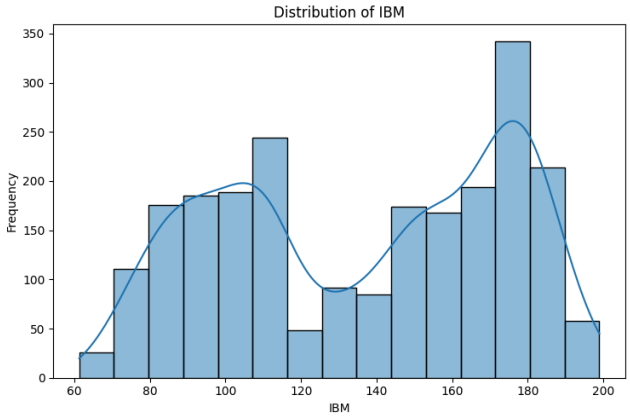

---Stats for SBUX---
 Mean: 23.213089442758022
 Median: 17.896952499999998
 Mode: [4.421114, 4.576884, 8.475711, 9.149186]
 Standard Deviation: 15.522634058857422
 Variance: 240.95216812520047
 Skewness: 0.8361669202826096
 Kurtosis: -0.2839467576607704

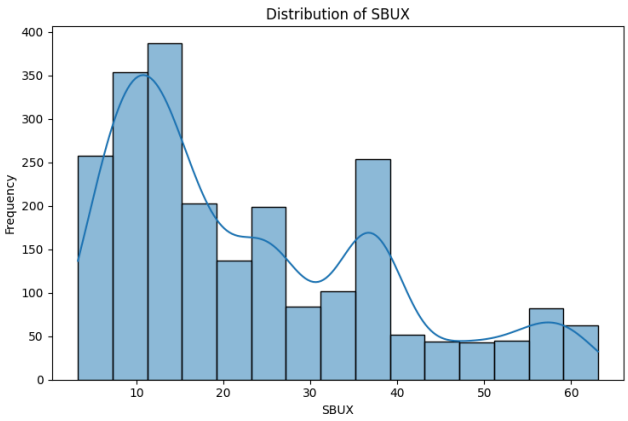

---Stats for AAPL---
 Mean: 54.71981027276669
 Median: 49.3274845
 Mode: [16.391781]
 Standard Deviation: 34.645281823743694
 Variance: 1200.2955526466253
 Skewness: 0.5172608251299968
 Kurtosis: -0.9016618135051773

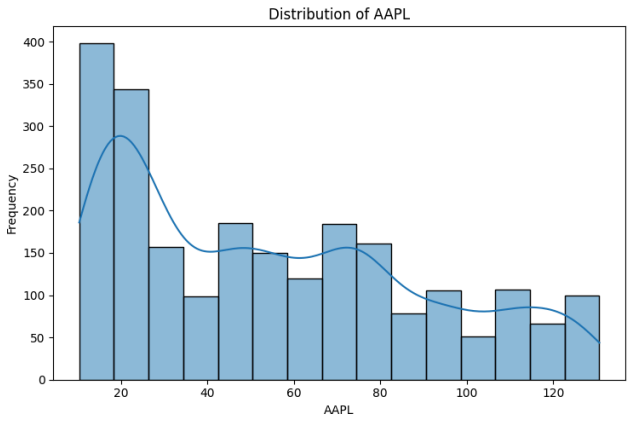

---Stats for GSPC---
 Mean: 1459.9998694236774
 Median: 1402.3700565
 Mode: [1064.880005, 1099.22998, 1178.099976, 1182.449951, 1271.5, 1343.800049, 1348.859985, 1360.030029, 1387.170044, 1416.
 Standard Deviation: 361.673515822571
 Variance: 130807.73204745952
 Skewness: 0.2650502148137591
 Kurtosis: -0.8511923676815929
```

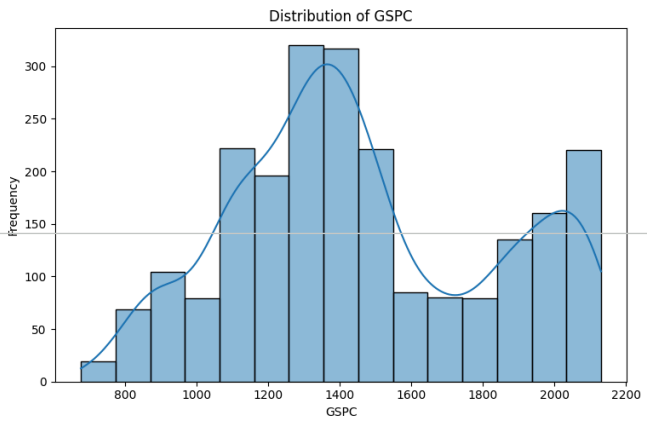
```
plt.figure(figsize=(15, 15))
for i, col in enumerate(numerical_cols):
    plt.subplot(3, 2, i + 1)
    sns.histplot(stock[col], kde=True, bins=15)
    plt.title(f'Distribution of {col}')
    plt.xlabel(col)
    plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```


Distribution of MSFT
Frequency
MSFT


Distribution of IBM
Frequency
IBM


Distribution of SBUX
Frequency
SBUX


Distribution of AAPL
Frequency
AAPL


Distribution of GSPC
Frequency
GSPC